

High-fidelity pixel-super-resolved lensless on-chip microscopy via height scanning and synthetic aperture: supplement

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High-fidelity pixel-super-resolved lensless on-chip microscopy via height scanning and synthetic aperture: supplemental document

This document provides supplementary information for “High-fidelity pixel-super-resolved lensless on-chip microscopy via height scanning and synthetic aperture”.

A. Supplementary Note 1. Numerical aperture analysis of the lensless on-chip microscopy system.

The effective numerical aperture (NA) in a lensless on-chip microscopy (LOCM) system is given by:

$$\text{NA}_{\text{eff}} = \text{NA}_{\text{illu}} + \text{NA}_{\text{dete}}, \quad (\text{S1})$$

where NA_{eff} denotes the effective NA, NA_{illu} denotes the illumination NA, and NA_{dete} denotes the detection NA. By introducing oblique incidence through multi-angle illumination, NA_{illu} can be increased to improve the system's NA_{eff} . For transmission-mode illumination, with an air refractive index $n \approx 1$, the maximum NA_{illu} is dictated by the illumination configuration:

$$\text{NA}_{\text{illu}} = \frac{D}{\sqrt{D^2 + L^2}}, \quad (\text{S2})$$

where D denotes the lateral displacement of the pinhole, and L denotes the vertical distance from the pinhole to the sample. NA_{illu} determines the extent of the sample spectrum shift. As a result, high spatial frequencies that were previously beyond the sensor's direct detection bandwidth can be translated into the detectable range through spectrum shifting by angled illumination. Integrating the shifted spectrum components from multiple illumination angles enables synthetic aperture expansion and more comprehensive retrieval of high-frequency details. The height-scanning LOCM system employs nearly vertical illumination, leading to $\text{NA}_{\text{illu}} \approx 0$. The maximum NA_{dete} is determined by the configuration of the detection system:

$$\text{NA}_{\text{dete}} = \frac{R}{\sqrt{R^2 + d_n^2}}, \quad (\text{S3})$$

where R denotes the effective radius of the CMOS sensor, and d_n represents the distance from the sample to the CMOS sensor.

Theoretically, the maximum NA_{illu} equals NA_{dete} . In practice, the maximum tolerable NA_{illu} must account for maintaining the integrity of the imaging region within the detector and the energy attenuation of the light source over propagation distance. In our experimental setup, $L \approx 70$ cm and $D \approx 10$ cm, corresponding to a maximum illumination angle of 8° from vertical, yielding a calculated $\text{NA}_{\text{illu}} = 0.14$. A CMOS sensor with 4608×3456 pixels and a pixel pitch of $1.34 \mu\text{m}$ yields an effective radius of $R = 3.86$ mm. Under the nearest sample-to-sensor distance of $d_n = 0.975$ mm, the resulting detection numerical aperture $\text{NA}_{\text{dete}} = 0.97$. This demonstrates that at the minimum imaging distance, the sensor covers an incident semi-angle of approximately 75.8° , offering high spatial frequency detection capability. The NA_{eff} of our LOCM system is 1.11, leading to a theoretical resolution limit [1, 2] $r = \lambda / \text{NA}_{\text{eff}} = 0.48 \mu\text{m}$. In the experimental results shown in the main text, the minimum resolvable line width observed is $1.10 \mu\text{m}$. When compared to the theoretical resolution limit, the actual resolution is found to be lower. Although a discrepancy exists, our reconstruction exceeds the limitation set by the sensor's pixel pitch ($< 1.34 \mu\text{m}$), indicating that the pixel super-resolution algorithm has overcome the Nyquist-Shannon sampling limit. Compared with in-line LOCM systems, our system achieves a 26% improvement in resolution.

The synthesized $\text{NA}_{\text{eff}} = 1.11$ represents an ideal combination assuming full spectral coverage. In practice, only three frames per axial height are used for synthetic aperture reconstruction, making it difficult to reach the theoretical NA limit. Furthermore, reconstruction of details near

or beyond the Nyquist-Shannon sampling limit is affected by environmental noise and pixel binning, which reduce the practical resolution.

B. Supplementary Note 2. Numerical simulation of quantitative phase reconstruction.

To validate the accuracy of phase reconstruction, a simulation of phase reconstruction was performed. The simulated reconstruction results for the “ship” phase image are shown in Fig. S1(a2)–(a4). All the algorithms can successfully reconstruct the sample image. In Fig. S1(a2), the lensless imaging using a synthetic aperture (LISA) [3] method suffers from limited diversity in the optical transfer function, resulting in a relatively higher mean-square error (MSE). The reconstruction results of multi-frame alternating projection (MFAP) [4] and height scanning synthetic aperture (HSSA) algorithm, shown in Fig. S1(a3) and (a4), respectively, both achieve low MSE values. In the reconstruction result of MFAP, high-frequency information is partially lost, resulting in a slight blurring of the letters on the ship. HSSA effectively compensates for this limitation. The simulated reconstruction results for the “mandrill” phase image are shown in Fig. S1(b2–b4), where all three methods achieve good phase reconstruction. Comparatively, HSSA achieves the smallest MSE, offering the best reconstruction performance.

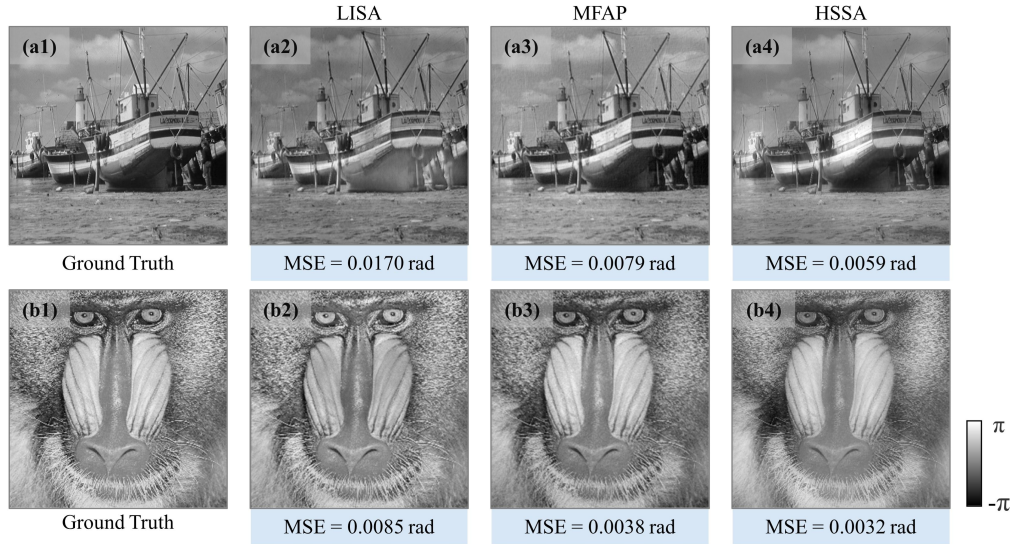


Fig. S1. Comparison of simulated reconstruction results for phase images. (a1) “Ship” image. (a2)–(a4) Phase reconstruction results of LISA, MFAP, and HSSA, respectively. (b1) “Mandrill” image. (b2)–(b4) Phase reconstruction results of LISA, MFAP, and HSSA, respectively.

C. Supplementary Note 3. HSSA validation on intricate structured samples.

To further verify the robustness of HSSA in pixel-super-resolved imaging, the experiment on the pine stem cross-section was conducted. The sample features intricate and non-uniform microscopic structures, which make phase retrieval and twin-image suppression more demanding.

The first captured hologram is shown in Fig. S2(a). The reconstruction results of LISA and MFAP are shown in Fig. S2(b) and Fig. S2(c), respectively. In the outer periderm region, both reconstructions suffer from twin-image artifacts. Fig. S2(d) presents the reconstruction result of HSSA, in which twin-image artifacts are substantially eliminated. Two regions of interest (ROIs) with distinct features were selected, as illustrated in Figs. S2(a1) and (a2). Comparative analysis of the ROI reconstructions confirms that HSSA robustly reconstructs the intricate structural details of the pith and xylem, providing additional evidence of its high fidelity and stability.

D. Supplementary Note 4. Time required for image sequence capture and reconstruction.

To evaluate the potential of HSSA in practical imaging applications, both the image sequence acquisition time and the reconstruction time were analyzed. In the experiment, the image stack with nine holograms was captured within 30 s. Our reconstruction was performed on a desktop computer with an AMD Ryzen 7 5800H CPU@3.20 GHz and GeForce RTX 3060 Laptop GPU. For nine frames of 4608×3456 holograms processed over 50 iterations, the HSSA completed in 105.33

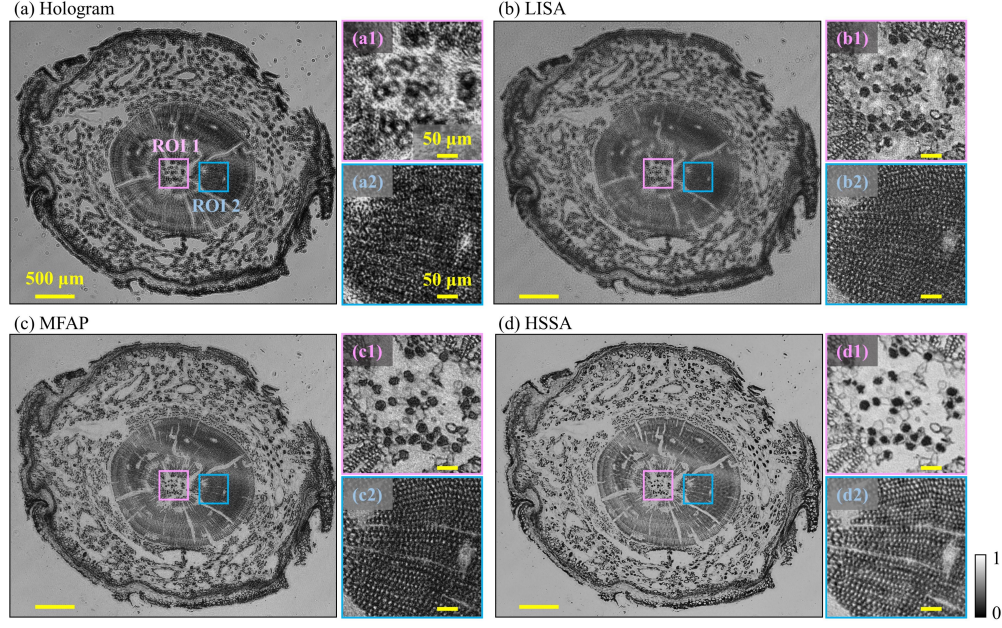


Fig. S2. Experimental reconstruction of the pine stem cross-section. (a) Raw captured hologram. (b)-(d) Amplitude reconstruction results by LISA, MFAP, and HSSA, respectively. (a1)-(d1) Magnified views of ROI 1. (a2)-(d2) Magnified views of ROI 2.

s. With the same number of input holograms, LISA required 110.01 s, and MFAP required 109.52 s. The reconstruction efficiency is comparable since the number of input image frames is the same. HSSA offers improved computational efficiency along with superior imaging fidelity.

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